Review plan · for the lecturer

Teaching AI agents at university: unit check

AI agents · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Agents and MDPs
- Planning and tools
- Evaluation
- Safety and governance

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you assess agent projects?

Exit ticket

- One change to my agents course this term, and how I will know it worked.

Try it in your next lecture
Bring your agent-project risk review and evaluation protocol to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **With γ = 0.9, a reward of 10 in two steps is worth now**

○ 9 ○ 10 ○ 8.1

2 **A course assistant is rewarded whenever a student rates its answer helpful. If the reward code has no bugs, the assistant will learn to give correct answers.**

○ True ○ False

3 **Estimate pass@k from samples.**

From n = 10 runs with c = 3 successes, the unbiased estimate of pass@2 is _____ .

Word bank: 0.51 · 0.60 · 0.53

4 **A* is guaranteed optimal when its heuristic is**

○ Large ○ Admissible ○ Random

5 **Order a ReAct step.**

☐ Observation

☐ Thought

☐ Answer

☐ Action

6 **Compute pass@k.**

With p = 0.5, pass@2 = _____ .

Word bank: 1 · 0.25 · 0.75

7 **Which improve evaluation validity? Choose all that apply.**

☐ Several runs ☐ Best-run reporting ☐ Held-out tasks

8 **On 50 tasks at 60%, the 95% interval is about**

○ ± 1 point ○ ± 14 points ○ ± 50 points

9 **Match each control to its risk.**

Least privilege	• • • •	Irreversible action
Approval gate	• • • •	Unclear accountability
Audit log	• • • •	Over-permissioned tool

10 **A grading agent can read submissions, draft feedback and publish final grades to the student records system. Which action most needs human approval?**

○ Publishing final grades ○ Reading submissions ○ Drafting feedback

Answers · for the lecturer

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1 **8.1**

A reward two steps away is discounted twice: $0.9^2 \times 10 = 0.81 \times 10 = 8.1$.

2 **False**

Ratings reward what pleases students, which is not always what is correct, so it can learn to tell them what they want to hear.

3 **From n = 10 runs with c = 3 successes, the unbiased estimate of pass@2 is 0.53.**

Two runs drawn from the ten both fail with probability $C(7,2)/C(10,2)$, so the estimate is $1 - 21/45 \approx 0.53$.

4 **Admissible**

An admissible heuristic never overestimates the remaining cost, so A* never passes over a cheaper path to the goal.

5 **Top to bottom: 3, 1, 4, 2**

Observe before answering.

6 **With p = 0.5, pass@2 = 0.75.**

Both tries fail with probability $(1 - 0.5)^2 = 0.25$, so pass@2 = 1 − 0.25 = 0.75.

7 **Held-out tasks; Several runs**

Held-out tasks and several runs keep the score honest; reporting only the best run picks the luckiest result.

8 **± 14 points**

The standard error is $\sqrt{(0.6 \times 0.4/50)} \approx 0.069$, and a 95% interval spans 1.96 standard errors each way: $1.96 \times 0.069 \approx 0.14$, about ±14 points.

9 **Least privilege — Over-permissioned tool**
Approval gate — Irreversible action
Audit log — Unclear accountability

Each control stops a different failure: least privilege limits what a tool can reach, an approval gate stops an irreversible action, and an audit log shows who did what.

10 **Publishing final grades**

Published grades reach every student's record and are hard to undo; reading and drafting change nothing until someone acts on them.